Chapter 1

Transforming Nonlinear Data with Logarithms

Certain relationships between two variables do not run in a straight line. Populations grow exponentially. Radioactive substances decay exponentially. When you try to fit a straight line to data like this, you may get a regression line, but the *line of best fit* will be wrong or non-existent. Logarithm models can straighten out curved relationships, allowing you to use linear regression techniques on data that is not linear in its raw form.

1.1 Exponential Models

A content creator posts a video and tracks how many views it gets each day:

Day (x)	0	1	2	3	4	5	6
Views, thousands (y)	4.2	8.3	17.4	34.1	77.8	161.8	337.5

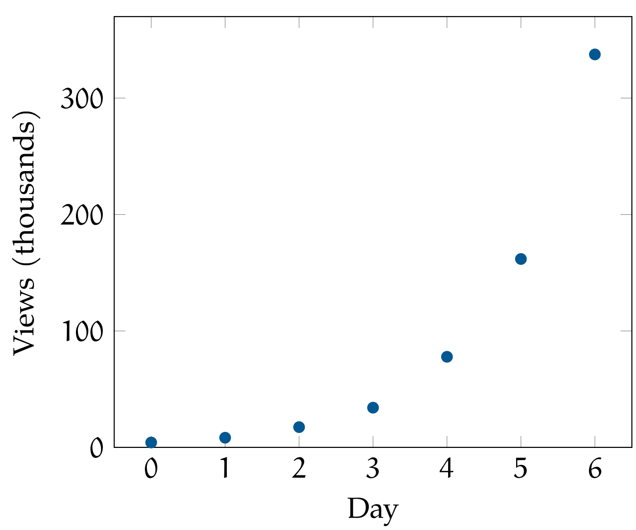

Figure 1.1: Daily views after posting. The curve bends upward instead of running straight.

A straight line fit to this data anyway would give $r^2 \approx 0.75$. That is not high enough to trust, and the residuals confirm it: the line misses low on both ends and high through the middle, a curved pattern rather than random scatter.

This would follow an exponential model, $y = a \cdot b^x$, where a is the starting count and b is the daily growth factor. Therefore:

$$\log(y) = \log(a \cdot b^x) = \log(a) + x\log(b)$$

$\log(a)$ and $\log(b)$ are both constants, so the right side is linear in x, with slope $\log(b)$ and intercept $\log(a)$. Graph $\log(y)$ against x instead of y against x, and a genuine exponential relationship becomes a line. This is a *logarithmic transformation*: the data does not change, only the scale you graph it on.

Apply that here:

Day (x)	0	1	2	3	4	5	6
Views, thousands (y)	4.2	8.3	17.4	34.1	77.8	161.8	337.5
$\log_{10}(y)$	0.623	0.919	1.241	1.533	1.891	2.209	2.528

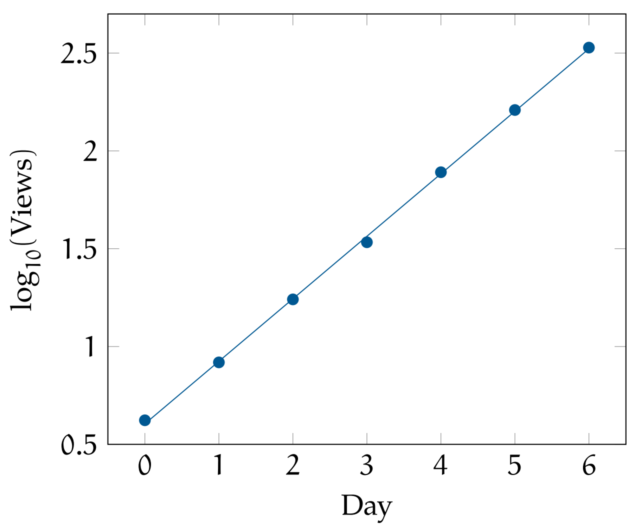

Figure 1.2: The same data after a logarithmic transformation. The points now fall along a line.

Regressing $\log_{10}(y)$ on x gives $\log_{10}(\hat{y}) = 0.3195x+0.6050$, with $r = 0.9997$ and $r^2 = 0.9995$.

Slope $= \log_{10}(b) = 0.3195$, so $b = 10^{0.3195} \approx 2.087$. Intercept $= \log_{10}(a) = 0.6050$, so $a = 10^{0.6050} \approx 4.027$. That gives:

$$\hat{y} = 4.027 \cdot (2.087)^x$$

To predict, work in the transformed scale first, then convert back. At $x = 9$: $\log_{10}(\hat{y}) = 0.3195(9) + 0.6050 = 3.481$, so $\hat{y} \approx 10^{3.481} \approx 3{,}022$ thousand views.

- If $y = a \cdot b^x$, then $\log(y) = \log(a) + x\log(b)$, linear in x.
- Graphing $\log(y)$ against x turns a genuine exponential relationship into a line.
- The transformed slope equals $\log(b)$; the transformed intercept equals $\log(a)$.
- To predict y, find $\log(\hat{y})$ from the linear model first, then raise 10 to that power.

Computing in Python

Here is the video views example again, this time in Python:

```
import numpy as np
import statsmodels.api as sm

day = np.array([0, 1, 2, 3, 4, 5, 6])
views = np.array([4.2, 8.3, 17.4, 34.1, 77.8, 161.8, 337.5])

# Fit the raw model: views as a linear function of day
X = sm.add_constant(day)
raw_model = sm.OLS(views, X).fit()
print(raw_model.params)
print("R-squared:", raw_model.rsquared)
print(raw_model.resid)
```

This gives an intercept of -54.91, a slope of 48.83, and $R^2 \approx 0.750$.

Apply the log transform and refit:

```
# Apply a natural log transform to the response variable
log_views = np.log(views)

log_model = sm.OLS(log_views, X).fit()
print(log_model.params)
print("R-squared:", log_model.rsquared)
print(log_model.resid)
```

Now the intercept is near 1.393, the slope near 0.7356, and $R^2 \approx 0.9995$.

One detail changes between the by-hand work and the code: `np.log()` is the natural logarithm, not $\log_{10}$. Natural log pairs with `np.exp()` instead, so the transformed slope and intercept convert back a little differently:

```
slope = log_model.params[1]
intercept = log_model.params[0]

```

```
growth_factor = np.exp(slope)
percent_growth = (growth_factor - 1) * 100
starting_views = np.exp(intercept)

print(f"Growth factor per day: {growth_factor:.3f}")
print(f"Percent growth per day: {percent_growth:.1f}%")
print(f"Estimated starting views: {starting_views:.1f} thousand")
```

This gives a growth factor of about 2.087, a growth rate of 108.7 percent per day, and 4,027 thousand views.

Exercise 1 Medication Concentration

Work

A patient takes a single dose of medication. A nurse measures the concentration remaining in the bloodstream every hour:

Hours (x)	Concentration, mg/L (y)
0	80.9
1	65.1
2	57.1
3	43.8
4	36.3
5	30.8
6	23.8

1. Find $\log_{10}(y)$ for each hour.
2. Regressing $\log_{10}(y)$ on x gives $\log_{10}(\hat{y}) = -0.0872x + 1.9108$, with $r = -0.9979$. Use this to write the exponential model $\hat{y} = a \cdot b^x$.
3. Predict the concentration at $x = 9$ hours. Is this interpolation or extrapolation?

Answer on Page 9

1.2 Power Models

Unlike exponential growth, certain curved relationships do not grow by a constant factor at every step. A marine biologist measures the length and weight of several fish from the same species:

Length, cm (x)	10	15	20	25	30	35	40
Weight, g (y)	15.7	53.9	135.5	244.0	461.4	668.0	1068.6

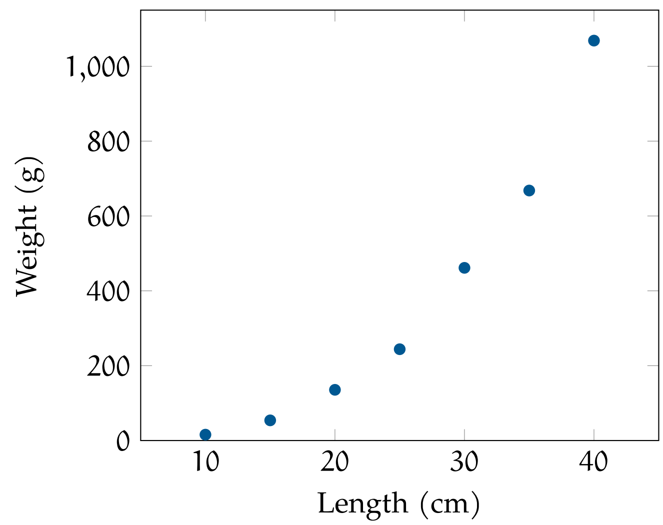

Figure 1.3: Fish weight against length. The curve bends upward, following a growth pattern distinct from exponential growth.

A straight line fit to this data yields $r^2 \approx 0.900$. Graphing $\log(y)$ against x, the same fix used in the last section, straightens the curve only slightly and leaves a pattern in the residuals. This relationship does not grow by a constant factor at every step the way the exponential model does; it grows by a constant power of x instead.

This kind of relationship follows a power model, $y = a \cdot x^b$. Take the logarithm of both sides:

$$\log(y) = \log(a \cdot x^b) = \log(a) + b\log(x)$$

Apply that to the fish data:

Length, cm (x)	10	15	20	25	30	35	40
Weight, g (y)	15.7	53.9	135.5	244.0	461.4	668.0	1068.6
$\log_{10}(x)$	1.000	1.176	1.301	1.398	1.477	1.544	1.602
$\log_{10}(y)$	1.196	1.732	2.132	2.387	2.664	2.825	3.029

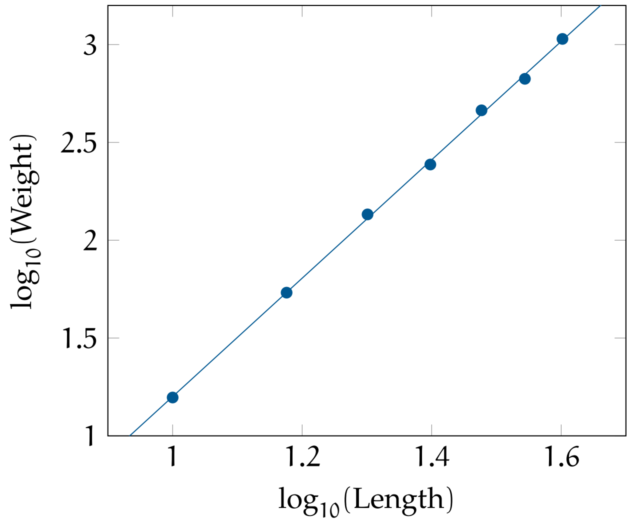

Figure 1.4: The fish data after a log-log transformation. Both axes are logarithms, and the points fall along a line.

Regressing $\log_{10}(y)$ on $\log_{10}(x)$ gives $\log_{10}(\hat{y}) = 3.0258 \log_{10}(x) - 1.8251$, with $r = 0.9997$ and $r^2 = 0.9993$, a much tighter fit than the raw linear attempt.

The slope is b itself, so $b \approx 3.026$. The intercept is $\log_{10}(a)$, so $a = 10^{-1.8251} \approx 0.01496$. That gives:

$$\hat{y} = 0.01496 \cdot x^{3.026}$$

An exponent close to 3 reflects real biology: fish grow in three dimensions at once, so weight scales close to the cube of length.

At $x = 45$ cm: $\log_{10}(\hat{y}) = 3.0258 \log_{10}(45) - 1.8251 = 3.177$, so $\hat{y} \approx 10^{3.177} \approx 1{,}503.9$ g.

- If $y = a \cdot x^b$, then $\log(y) = \log(a) + b \log(x)$, linear in $\log(x)$.
- Graphing $\log(y)$ against $\log(x)$, a *log-log plot*, turns a genuine power relationship into a line.
- The transformed slope equals b directly; the transformed intercept equals $\log(a)$.
- To predict y, find $\log(x)$ first, use the linear model to get $\log(\hat{y})$, then raise 10 to that power.

Exercise 2 Fall Time

A student drops a ball from different heights and times how long it takes to hit the ground:

Height, m (x)	Fall time, s (y)
5	1.01
10	1.42
15	1.71
20	2.02
25	2.24
30	2.45
35	2.59

1. Explain why the shape of this scatterplot points to a power model rather than an exponential one.
2. Regressing $\log_{10}(y)$ on $\log_{10}(x)$ gives $\log_{10}(\hat{y}) = 0.4908\log_{10}(x) - 0.3388$, with $r = 0.9996$. Use this to write the power model $\hat{y} = a \cdot x^b$.
3. Predict the fall time from a height of 50 m.

Answer on Page 9

1.3 Choosing the Right Transformation

Both sections in this chapter start from the same move: something is curved, so take a logarithm and see if it straightens out. The two models call for a different logarithm, and mixing them up wastes time on a transformation that will not work.

Model	Equation	Transform	Linear in
Exponential	$y = a \cdot b^x$	$\log(y)$ vs. x	x
Power	$y = a \cdot x^b$	$\log(y)$ vs. $\log(x)$	$\log(x)$

When you are unsure which model fits, try both transformations and compare. Graph $\log(y)$ against x: if that looks straight, the relationship is exponential. If it still curves, graph $\log(y)$ against $\log(x)$ instead; if that straightens out, it is a power model. Whichever transformation gives the higher r^2 with no leftover pattern in the residuals is the one that matches the data.

These two transformations show up often outside the classroom: population growth tends to follow an exponential pattern, while relationships like body mass and metabolic rate, or a planet's distance from the sun and its orbital period, tend to follow a power model.

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 4)

Hours (x)	0	1	2	3	4	5	6
$\log_{10}(y)$	1.908	1.814	1.757	1.641	1.560	1.489	1.377

Slope $= \log_{10}(b) = -0.0872$, so $b = 10^{-0.0872} \approx 0.818$. Intercept $= \log_{10}(a) = 1.9108$, so $a = 10^{1.9108} \approx 81.43$. The model is:

$$\hat{y} = 81.43 \cdot (0.818)^x$$

The concentration starts around 81.43 mg/L and multiplies by about 0.818 each hour, dropping roughly 18 percent per hour.

At $x = 9$: $\log_{10}(\hat{y}) = -0.0872(9) + 1.9108 = 1.126$, so $\hat{y} \approx 10^{1.126} \approx 13.37$ mg/L. The data only covers hours 0 through 6, so $x = 9$ is extrapolation.

Answer to Exercise 2 (on page 7)

The raw scatterplot rises quickly at first and then levels off, growing more slowly as height increases. Exponential growth does the opposite: it starts slow and accelerates. That mismatch in shape is the signal to try a power model instead.

Slope $= b \approx 0.4908$. Intercept $= \log_{10}(a) = -0.3388$, so $a = 10^{-0.3388} \approx 0.4584$. The model is:

$$\hat{y} = 0.4584 \cdot x^{0.4908}$$

An exponent near 0.5 matches the physics: fall time under gravity is proportional to the square root of height, $t \propto \sqrt{h}$.

At $x = 50$: $\log_{10}(50) \approx 1.699$, so $\log_{10}(\hat{y}) = 0.4908(1.699) - 0.3388 = 0.495$, giving $\hat{y} \approx 10^{0.495} \approx 3.13$ s.

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